08. Model Evaluation

Interpret the recorded metrics and the limits of the available test partitions.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Baseline Random Forest (main notebook)

Partition	Accuracy	Important observation
Validation	82.54%	Stunting F1=0.91; Overweight F1=0.22; Underweight/Wasting F1=0
Test	72.22%	Stunting F1=0.85; Overweight F1=0; Underweight F1=0; Wasting absent

Linear SVM (main notebook)

Partition	Accuracy	Important observation
Validation	91.27%	Stunting recall=0.99; Underweight and Wasting recall=0
Test	84.13%	Stunting F1=0.91; Overweight F1=0.61; Underweight and Wasting F1=0

SMOTE + Random Forest (separate SMOTE notebook)

Partition	Accuracy	Macro F1	Weighted F1
Validation	74%	0.25	0.76
Test	64%	Not fully reliable from truncated output	Not fully reliable from truncated output

SMOTE + linear SVM / tuned SVM

The unscaled linear SVM after SMOTE reports very poor performance (about 20% accuracy). After scaling and GridSearchCV, the tuned SVM reports 55% validation accuracy and 48% test accuracy in the saved output. This illustrates why scaling matters for SVM, but the result remains weak.

How to interpret

- High accuracy largely reflects the majority Stunting class.
- The confusion matrices show the models frequently predict Stunting for minority-class records.
- UndefinedMetricWarning appears when a class has no predicted examples; precision/F1 are set to zero by the metric implementation.
- The test split has no Wasting examples, so four-class test evaluation is incomplete.
- Metrics from different notebooks are not directly comparable because feature sets, sampling procedures and model configurations differ.

Metric definitions

- Accuracy = correct predictions / all predictions.
- Per-class precision = TP/(TP+FP); of the records predicted as a class, how many truly belong to it?

- Per-class recall = TP/(TP+FN); of the true class records, how many were detected?
- F1 = harmonic mean of precision and recall.
- Macro-F1 averages class F1 equally, giving rare classes equal importance; weighted-F1 weights by class support.

No defensible winner

The SVM's 84.13% test accuracy is higher than the baseline Random Forest's 72.22%, but neither result demonstrates reliable four-class recognition. Wasting is absent from the test set and minority-class F1 is poor. Do not present 84.13% as a clinically meaningful model success.